Homework 6: Greeks, combinations, and implied volatility

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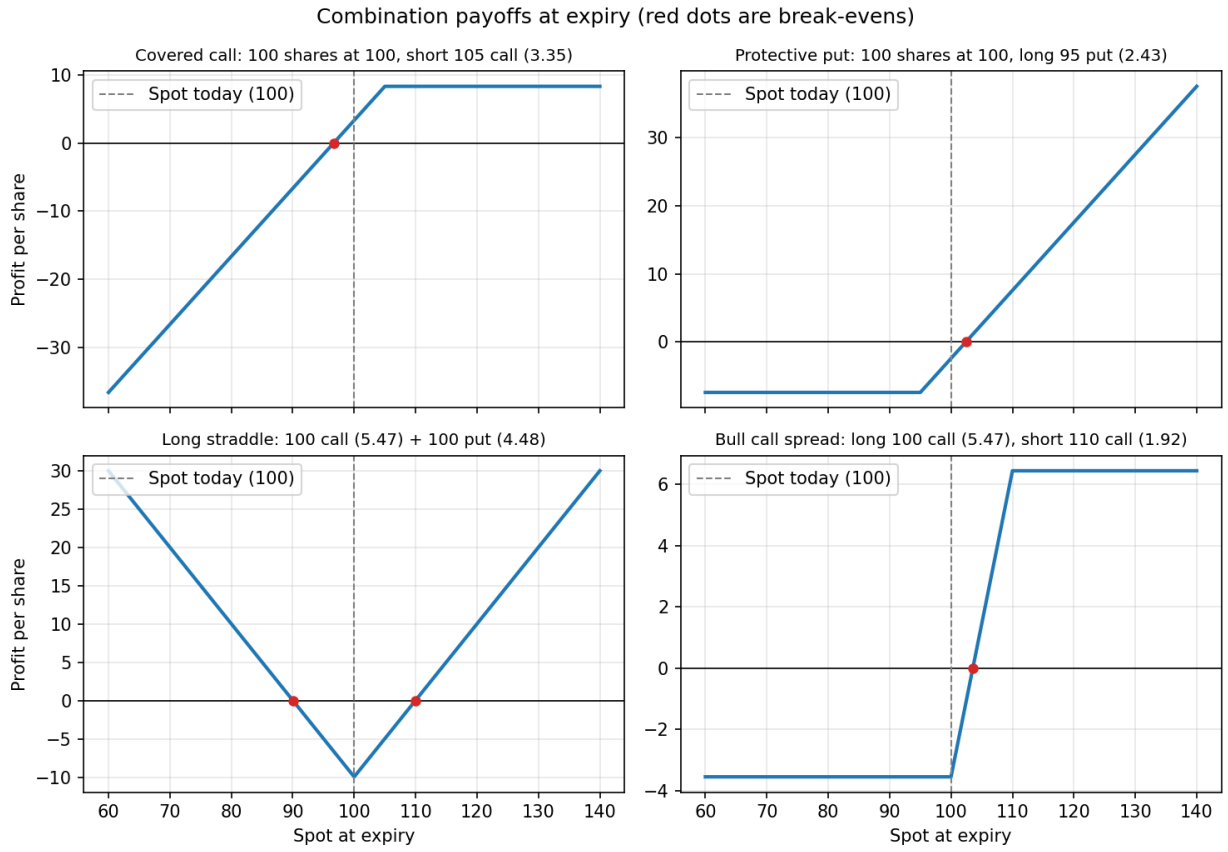
Combinations

1. Four positions: max profit, max loss, break-evens

A hypothetical stock at 100, with every premium priced by my Black-Scholes function (three months to expiry, 4% rate, 25% volatility) and rounded to the cent, so the numbers hang together. Each position is built in HW6.py by adding the Homework 5 `payoff` function to itself or to a share position (`covered_call` keeps the starter's name). Max profit, max loss and break-evens are read off a 0.01 grid of expiry prices from 0 to 1000; a profit still rising at the top of the grid is reported as unlimited.

```
spot 100; premiums are Black-Scholes, T = 0.25 years, r = 4%, vol = 25%
calls: 100 = 5.47, 105 = 3.35, 110 = 1.92    puts: 95 = 2.43, 100 = 4.48
per share (x100 for one contract)
position            max profit    max loss  break-evens
Covered call              8.35      -96.65  96.65
Protective put       unlimited       -7.43  102.43
Long straddle        unlimited       -9.95  90.05, 109.95
Bull call spread          6.45       -3.55  103.55
```

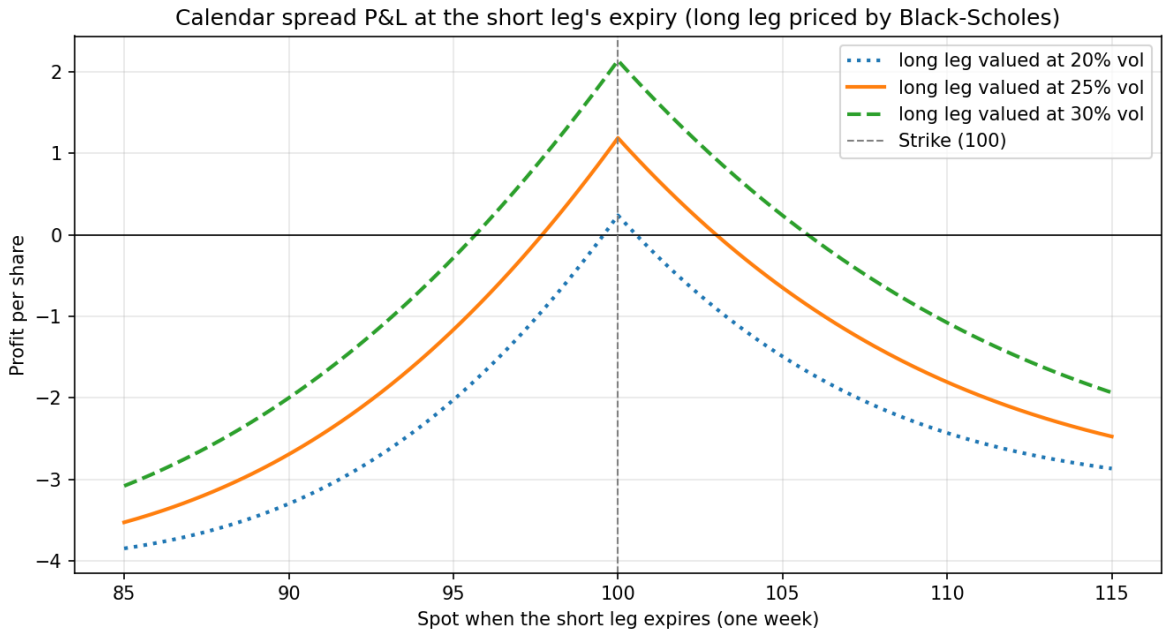
Every figure is also asserted in HW6.py against the algebra. **Covered call**: the most it can make is the 5 of stock gain up to the strike plus the 3.35 premium, 8.35, reached anywhere above 105; the premium only cushions the downside, so the stock going to zero costs 96.65, and break-even is 100 - 3.35. **Protective put**: the put floors the loss at 5 of stock loss plus the 2.43 premium, 7.43, anywhere below 95; upside is unlimited and break-even is 100 + 2.43. **Long straddle**: the worst case is spot finishing exactly at 100, losing both premiums, 9.95; it profits outside 100 ± 9.95. **Bull call spread**: a 3.55 net debit buys a 10-wide spread, so the most it can make is 6.45 above 110, and break-even is 100 + 3.55. One contract is 100 shares, so multiply every figure by 100 (the covered call risks $9,665 to make $835).



2. Calendar spread

```
calendar spread, strike 100: buy 3-month call 5.47, sell 1-week call 1.42, net debit 4.05
one week later, underlying unchanged at 100:
  short leg expires worthless: +1.42
  long leg now worth 5.24, lost 0.23
  spread P&L +1.19 per share
```

An expiry payoff diagram works because on the expiry date every leg is worth exactly its intrinsic value, which depends only on spot. A calendar has no such date: when the one-week call expires, the three-month call still has twelve weeks left, and its value depends on spot, the time remaining and whatever implied volatility is at that moment, so the position is a family of curves, not one line (the plot below values the long leg with Black-Scholes at three volatilities). What the position is betting on is time decay and stillness: if the underlying does nothing for a week the short call decays all the way from 1.42 to zero while the long call only loses 0.23, so the spread makes 1.19 on a 4.05 debit. It loses if the stock moves far either way (both calls converge to the same intrinsic value, wiping out the time value the long leg was bought for), and because the leg left standing is the long one, it gains if implied volatility rises and loses if it falls.



Black-Scholes and implied volatility

3. Black-Scholes price

```
S=100, K=100, T=1, r=4%, vol=25%
call 11.8370   put 7.9160
C - P = 3.9211,  S - K*exp(-rT) = 3.9211
```

Both match the check values (11.84 and 7.92), and put-call parity holds to four decimals. The normal CDF is built from `math.erf`, as the starter imports.

4. implied_volatility

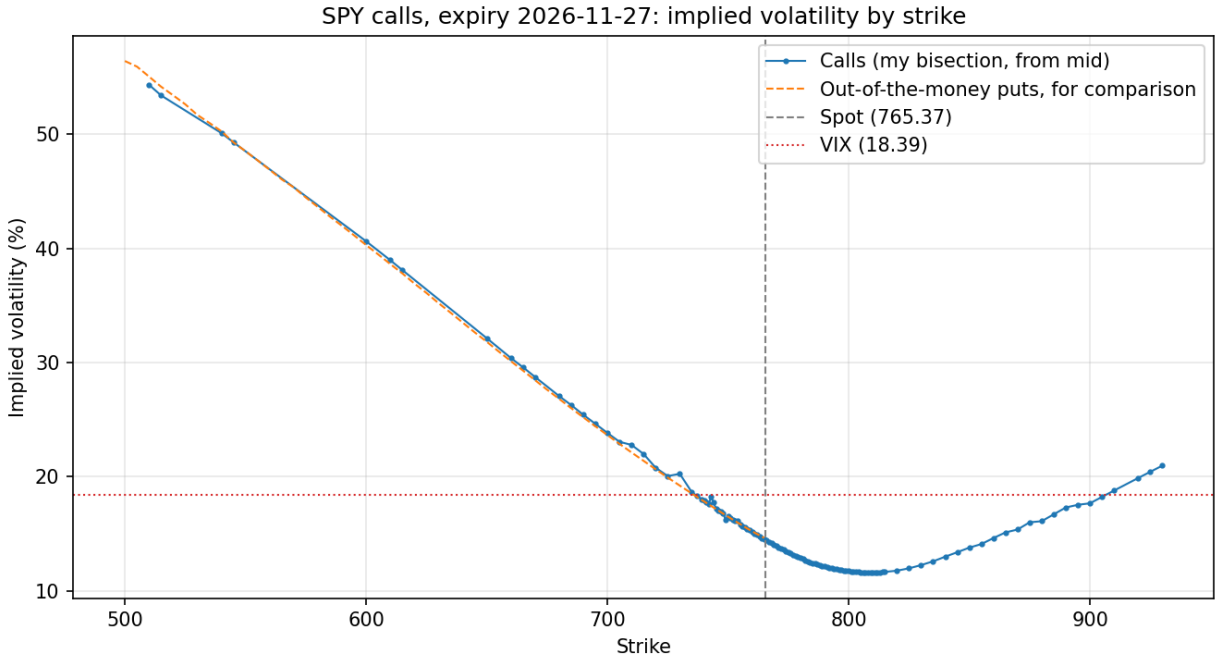
```
feed Black-Scholes prices back into implied_volatility
call K=100  vol 0.2500 -> price  11.8370 -> implied 0.2500
put  K=100  vol 0.2500 -> price   7.9160 -> implied 0.2500
put  K=100  vol 0.4000 -> price  13.6572 -> implied 0.4000
call K=120  vol 0.1800 -> price   2.3561 -> implied 0.1800
```

Bisection between 0.0001 and 5.0, halving until the bracket is narrower than 1e-8 (about 29 steps). It works because price rises monotonically with volatility. If a quote lies outside the prices reachable in that range, for instance below the no-arbitrage floor, it returns nan instead of a wrong number. It recovers 0.2500, 0.4000 and 0.1800 exactly, on calls and puts and out of the money. check_hw06.py passes all 10 of its checks.

5. Implied volatility across a live chain

SPY options from yfinance, pulled Monday 26 October 2026 after the close, so spot (765.37) and the quotes are that day's closing values. The expiry is the one closest to 30 days out; time is calendar days / 365, the rate is the 13-week T-bill, and implied volatility is solved from the mid. No dividend adjustment is needed because SPY's next ex-dividend date (mid-December) is after this expiry. The quotes used are saved as chain_2026-11-27.csv.

```
SPY calls, expiry 2026-11-27 (32 days), spot 765.37, rate 4.03% (13-week T-bill)
123 calls, 0 zero bid, 0 with a mid below the no-arbitrage floor (no volatility fits), 123 solved
implied vol at strike 765.0 (nearest spot): 14.53%  (put at the same strike: 14.65%; yfinance's own
    call figure: 16.12%)
VIX today: 18.39  (17.1 on 2026-09-14)
  strike  % of spot       mid  implied vol
   610.0      79.7%   158.190       38.96%
   650.0      84.9%   118.730       32.13%
   690.0      90.2%    79.700       25.43%
   725.0      94.7%    46.805       20.06%
   765.0     100.0%    14.695       14.53%
   785.0     102.6%     4.890       12.46%
   804.0     105.0%     1.175       11.68%
   825.0     107.8%     0.230       11.99%
lowest implied vol 11.62% at strike 810.0; highest 54.31% at strike 510.0
```



The shape is the **volatility skew**, also called the smirk: the equity-index form of the volatility smile. Implied volatility is 39% at 80% of spot, 14.5% at the money, bottoms at 11.6% around 106% of spot and then curls back up to about 21% at the highest strikes. If the market believed Black-Scholes, with normally distributed log returns and one volatility, every strike would give the same number and the line would be flat. A steep left side says the market prices large falls as far more likely than a normal distribution allows: a fat left tail and negative skewness, which is what crashes like 2008, 2020 and April 2025 look like. The smaller rise on the right says big rallies are also more likely than normal (fat tails on both sides), just less so than big falls. The skew also explains why VIX (18.39) sits above at-the-money vol (14.53%): VIX is computed from a strip of out-of-the-money options across all strikes, so it includes the expensive wings.

Two checks on the data. The deep in-the-money calls have little extrinsic value relative to their bid-ask spreads, so their implied vols are sensitive to the mid; but out-of-the-money puts at the same strikes, with spreads of a cent, land on the same curve, which is put-call parity holding. Second, yfinance's own `impliedVolatility` column is not reliable: it says 16.12% for the 765 call but 13.24% for the 765 put, while my solver gives 14.53% and 14.65%.

6. Realised against implied

```
30 trading days of log returns, 2026-09-15 to 2026-10-26
  realised vol   ATM implied     VIX     gap
        10.49%        14.53%  18.39%   4.03%
```

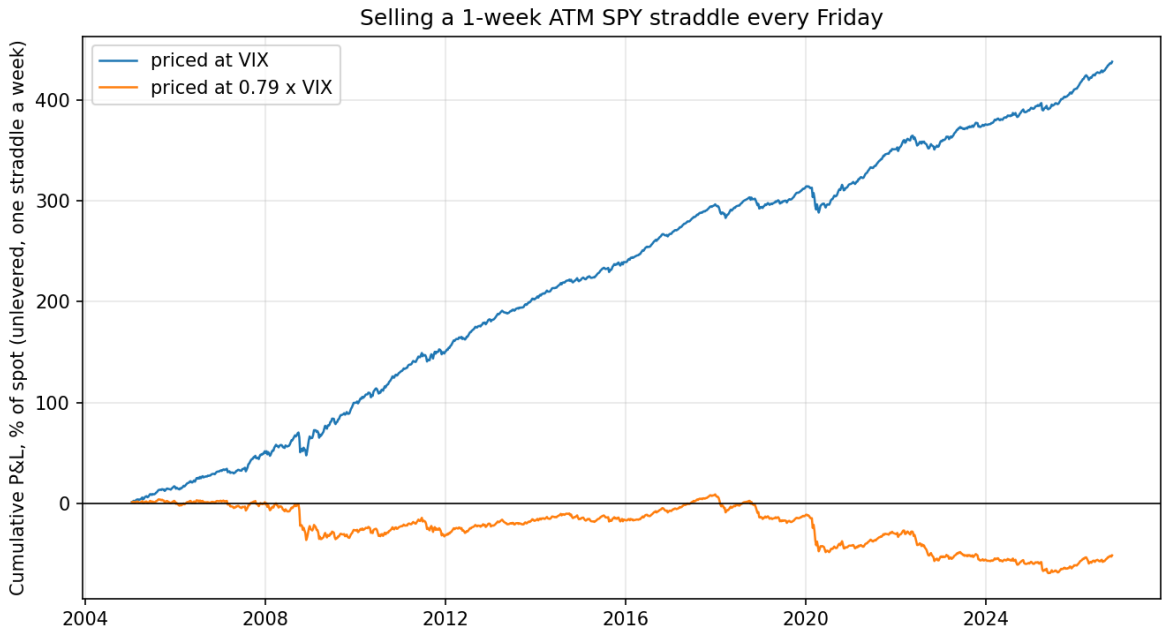
Realised volatility is the sample standard deviation of the last 30 daily log returns, times √252. At-the-money implied is 4.0 volatility points above it, and VIX is 7.9 points above.

Written

7. Implied against realised, and selling options every week

It holds for SPY: 14.53% implied at the money against 10.49% realised over the last 30 days. It is not a one-day accident either. Going back to 2005, VIX was above the volatility SPY actually delivered over the following month on 82% of Fridays, by a median of 4.2 points. To see what happens to someone who collects that gap, HW6.py sells a one-week at-the-money SPY straddle every Friday since 2005, priced by Black-Scholes, and settles it the next Friday. Because VIX overstates at-the-money vol (see above), it runs twice: priced at VIX, and at 0.79 x VIX (today's ATM / VIX ratio).

```
2005-01-07 to 2026-09-18, 1093 Fridays: VIX above the next month's realised vol 82% of the time, by
    4.17% at the median
selling a 1-week ATM straddle every Friday, priced at VIX, 1097 weeks, P&L as % of spot:
  wins 69% of weeks; average win 1.10%, average loss -1.17%; mean week +0.400%
  premium averages 2.13% of spot, which is the most any week can make
  cumulative +438.3% of spot; deepest drawdown -26.2%, bottoming in the week to 2020-04-17
  five worst weeks:
    week to 2008-10-10: premium 4.99%, P&L -14.81%, = 7.0 average weeks of premium
    week to 2020-02-28: premium 1.89%, P&L -9.27%, = 4.4 average weeks of premium
    week to 2020-04-17: premium 7.31%, P&L -8.18%, = 3.8 average weeks of premium
    week to 2020-03-20: premium 6.39%, P&L -8.16%, = 3.8 average weeks of premium
    week to 2025-04-04: premium 2.39%, P&L -6.67%, = 3.1 average weeks of premium
  sold on margin at L x capital, the worst week costs L x 14.81%: 3x -> -44%, 5x -> -74%, 7x -> -104%;
       the account is gone above 6.8x
selling a 1-week ATM straddle every Friday, priced at 0.79 x VIX (today's ATM / VIX ratio), 1097
    weeks, P&L as % of spot:
  wins 57% of weeks; average win 0.86%, average loss -1.24%; mean week -0.047%
  premium averages 1.68% of spot, which is the most any week can make
  cumulative -51.7% of spot; deepest drawdown -78.2%, bottoming in the week to 2025-05-23
  five worst weeks:
    week to 2008-10-10: premium 3.94%, P&L -15.85%, = 9.4 average weeks of premium
    week to 2020-04-17: premium 5.78%, P&L -9.72%, = 5.8 average weeks of premium
    week to 2020-02-28: premium 1.49%, P&L -9.67%, = 5.8 average weeks of premium
    week to 2020-03-20: premium 5.05%, P&L -9.50%, = 5.7 average weeks of premium
    week to 2025-04-04: premium 1.89%, P&L -7.18%, = 4.3 average weeks of premium
  sold on margin at L x capital, the worst week costs L x 15.85%: 3x -> -48%, 5x -> -79%, 7x -> -111%;
       the account is gone above 6.3x
```



A seller can reasonably demand the gap because they are selling insurance. Their best week is the premium, about 2% of spot; their worst is many times that, and it comes at the worst possible time: in a crash, when their other assets are falling, volatility is exploding and the broker is raising margin. The buyers, mostly portfolio managers hedging stock, value protection in exactly those states, and that crash risk cannot be diversified away because it hits everything at once. So, like any insurer, the seller charges more than the expected loss, and the gap between implied and realised is that charge. Part of the measured gap is also a sampling effect: implied volatility prices the crash that comes one month in twenty, and most 30-day windows of realised volatility, like this one, do not contain it.

The person who sells every week gets exactly what the backtest shows. They win most weeks, 69% of them at VIX prices, and the equity curve climbs smoothly for years, which is what makes the trade popular. But the wins are small and capped while the losses are not: the week to 10 October 2008 lost 14.8% of spot, seven average weeks of premium, and three of the five worst weeks came within two months of each other in 2020. Whether the whole thing makes money at all depends on a 20% difference in the premium: at VIX it earns, at 0.79 x VIX the same trades lose 52% of spot with a 78% drawdown. Because the unlevered return is small, sellers usually lever it up on margin, and that is how it ends. At five times capital the October 2008 week is a 74% loss; at seven times the account is gone, and a blown-up seller never collects the rich premiums that follow, which are exactly what pull the at-VIX curve back up. This is not hypothetical. On 5 February 2018 VIX went from 17.31 to 37.32 in one day, the short-volatility note XIV lost about 96% of its value and Credit Suisse shut it down, and LJM Partners, a fund that had sold S&P options profitably for years, lost around 80% that week. In November 2018 OptionSellers.com, which sold options on natural gas for retail clients, was wiped out by a spike in gas prices and its clients ended up owing their broker money on top of losing everything. The strategy makes money most weeks; it ends in a single week that takes everything it made and the capital underneath it.